Question 1: Object-Oriented Programming (OOP) [10 marks]

Introduction: (2 marks)

Object-Oriented Programming is a programming style that uses objects to organize code. Objects have data and methods. OOP makes programs easier to manage. Languages like Java, C++, and Python support OOP.

Core Concepts: (4 marks)

OOP has four main concepts. Encapsulation means hiding data inside a class using private variables and providing public methods to access them. Inheritance means a child class can use methods and properties from a parent class. This helps in reusing code. Polymorphism means the same method can behave differently in different classes. Abstraction hides complex details and shows only necessary information.

Examples: (2 marks)

A Car class can inherit from Vehicle class. Car has its own start() method that overrides Vehicle's start(). This shows inheritance and polymorphism. BankAccount class keeps balance private and has deposit and withdraw methods.

Conclusion: (2 marks)

OOP is useful for building large programs. It helps in code reuse and makes programs easier to understand. The four pillars make software development better.

Question 2: Database Normalization [10 marks]

Introduction: (2 marks)

Database normalization is the process of organizing database tables to reduce redundancy and dependency. It involves dividing tables into smaller tables. Normalization helps avoid problems when inserting, updating, or deleting data.

Core Concepts: (4 marks)

First Normal Form (1NF) means each column should have atomic values and no repeating groups. Second Normal Form (2NF) means the table should be in 1NF and all columns should depend on the full primary key. Third Normal Form (3NF) means no column should depend on a non-key column. These forms help make the database more organized.

Examples: (2 marks)

A student table with StudentID, StudentName, CourseID, CourseName has redundancy because CourseName repeats. We can split it into Student table and Course table to remove this redundancy.

Conclusion: (2 marks)

Normalization is important for good database design. It reduces redundancy and maintains data integrity. Higher normal forms give better structure to databases.

Question 3: Operating System Process Management [10 marks]

Introduction: (2 marks)

Process management is a function of the operating system that manages running programs. A process is a program that is currently running. The OS handles multiple processes at the same time by sharing CPU and memory resources.

Core Concepts: (4 marks)

A process has states: New, Ready, Running, Waiting, and Terminated. The OS uses a Process Control Block (PCB) to store information about each process like its ID, state, and registers. CPU scheduling decides which process runs next. FCFS runs processes in order of arrival. Round Robin gives each process a fixed time slice.

Examples: (2 marks)

In Round Robin scheduling with quantum 4ms, processes take turns. If three processes P1, P2, P3 are ready, they each get 4ms turns repeatedly until they finish.

Conclusion: (2 marks)

Process management ensures the CPU is used efficiently. Scheduling algorithms help balance performance and fairness. The OS must handle multiple processes running at the same time without conflicts.

Question 4: Computer Networks and the OSI Model [10 marks]

Introduction: (2 marks)

A computer network connects multiple computers so they can share data and resources. The OSI model is a standard model with seven layers that describes how data travels across a network from one device to another.

Core Concepts: (4 marks)

The OSI model has 7 layers. Application layer is the topmost layer used by applications like browsers. Transport layer uses TCP and UDP to send data reliably or quickly. Network layer handles IP addressing and routing using routers. Data Link layer handles MAC addresses and error detection. Physical layer transmits actual bits over cables or wireless signals.

Examples: (2 marks)

When we open a website, the browser sends an HTTP request. TCP breaks it into segments, IP adds addresses, and it travels over the network to reach the server. The server sends back the webpage the same way.

Conclusion: (2 marks)

The OSI model helps understand how networks work by dividing communication into layers. Each layer has a specific job. TCP/IP is the practical version used on the internet today.